Quadratic Forms and the Clifford Product

From a declared quadratic form to the product it specifies

Session 13a of three: 13b *Programs and Interpreters*, 13c *Wedge, Dual, and Binding*

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Roadmap

I	One vector in both slots	the quadratic form of a matrix; it sees only the symmetric half; polarization
II	Quadratic forms at work	Mahalanobis distance, label propagation, and their signatures
III	From a quadratic form to a product	declare Q, demand $\mathbf{v}\mathbf{v} = Q(\mathbf{v})$; both halves follow
IV	The smallest algebras	one letter: complex, split-complex, dual numbers; two: quaternions
V	Close	what 13b and 13c build on this; exercises; reading

Assumed, from the Signatures session, and recalled where needed: an inner product is a matrix in the middle; a declaration fixes each axis's square to $+1$, -1, or 0; polarization turns lengths into similarities; a signature is read off eigenvalues.

Added: a product of vectors specified by one rule, $\mathbf{v}\mathbf{v} = Q(\mathbf{v})$. Polarization forces its symmetric half; the antisymmetric half, the wedge, comes free.

Every number here is arithmetic you can check by hand.

Where We Are: The Attention Head of Session 11

An attention head can score the same pair of tokens differently depending on which token supplies the query. Two tokens, $\mathbf{x}_1 = (3, 1)$ and $\mathbf{x}_2 = (1, 3)$, and a learned $G = W_Q^\top W_K$; the **raw score** of query $\mathbf{x}$ against key $\mathbf{y}$ is $\mathbf{x}^\top G \mathbf{y}$, before scaling, masking and softmax:

label	query	key	raw score
$1 \to 2$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_1^\top G \mathbf{x}_2 = 24$
$2 \to 1$	$\mathbf{x}_2$	$\mathbf{x}_1$	$\mathbf{x}_2^\top G \mathbf{x}_1 = 8$

Split G into its symmetric and antisymmetric halves to see where the difference comes from:

$$G = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{S=\frac{1}{2}(G+G^\top)} + \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{A=\frac{1}{2}(G-G^\top)}$$

$\mathbf{x}^\top S \mathbf{y}$	16	16	symmetric compatibility
$\mathbf{x}^\top A \mathbf{y}$	$+8$	-8	directional contribution

So $24 = 16 + 8$ and $8 = 16 - 8$. The symmetric half contributes the same whichever token is the query; the antisymmetric half adds to one direction what it subtracts from the other. A token scored against itself gets no antisymmetric contribution, since $\mathbf{x}^\top A \mathbf{x} = 0$: both tokens have self-score $(x_1 + x_2)^2 = 16$. Session 11's product $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ keeps both halves, for the dot product only. Today: the same product for any declared form.

The Question for Today

A puzzle

The Signatures session showed that a declaration fixes a symmetric form and nothing else: the square of each axis, and by polarization every similarity, all of it blind to order. Yet the algebra built from the declaration keeps order: $\mathbf{a} \wedge \mathbf{b} = -\mathbf{b} \wedge \mathbf{a}$. Where does the order-sensitive half come from, if it was never declared?

The answer comes in three steps, one per part of the deck:

1. Put one vector into both slots of a bilinear form. What comes out is a **quadratic form**, and it sees only the symmetric half. (Part I)
2. A quadratic form is the symmetric half, no more and no less: **polarization** recovers it exactly, and nothing recovers the antisymmetric half. (Parts I, II)
3. Demand a product in which every vector squares to its quadratic form. Polarization then fixes the symmetric half of the product, and the antisymmetric half comes along unconstrained, as a new kind of object. (Part III)

So declaring Q is enough, because the algebra supplies the half that Q cannot see.

Part I
One Vector in Both Slots

The Matrix in the Middle, Recalled

Every inner product in the course has the shape $\mathbf{x}^\top M\mathbf{y}$, and the three kinds of M met so far:

source	matrix in the middle	example	symmetric?
assumed	I	the dot product, the cosine	yes
declared	D, diagonal with entries $+1$, -1, 0	the Signatures session	yes
learned	$G = W_Q^\top W_K$	the attention head	no: $G = S + A$

A matrix in the middle is called a **bilinear form**: a function $B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top M\mathbf{y}$ of two vectors, linear in each. The first two rows are symmetric, $B(\mathbf{x}, \mathbf{y}) = B(\mathbf{y}, \mathbf{x})$; the head is not, which is what the 24 and the 8 showed.

This part asks one question about every such M: what do you see if you put the same vector in both slots? The answer is the quadratic form, and it turns out to see exactly the symmetric half of M, whichever row M came from.

Feed the Same Vector to Both Slots

A bilinear form takes two vectors. Give it the same vector twice:

$$B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top G\, \mathbf{y} \qquad \rightsquigarrow \qquad Q(\mathbf{x}) \;=\; B(\mathbf{x}, \mathbf{x}) \;=\; \mathbf{x}^\top G\, \mathbf{x}$$

Q is the **quadratic form** of B: a function of one vector, quadratic in its coordinates. For the head, compute $G\mathbf{x}$ first and then dot with $\mathbf{x}$:

	$G\mathbf{x}$	$Q(\mathbf{x}) = \mathbf{x}^\top(G\mathbf{x})$
$\mathbf{x}_1 = (3, 1)$	$(5,\ 1)$	$3 \cdot 5 + 1 \cdot 1 = 16$
$\mathbf{x}_2 = (1, 3)$	$(7,\ 3)$	$1 \cdot 7 + 3 \cdot 3 = 16$

Written as a polynomial, $Q(\mathbf{x}) = x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2$: this head's self-score. Both tokens have squared length 16 under it.

You have met this already: $\mathbf{x}^\top I\mathbf{x} = \|\mathbf{x}\|^2$ is the quadratic form of the dot product, $\mathbf{x}^\top D\mathbf{x} = \sum_i (\mathbf{e}_i \cdot \mathbf{e}_i)\, x_i^2$ is the quadratic form of a declaration, and every squared length in the course has been one.

The Antisymmetric Half Disappears

For any antisymmetric A, the number $\mathbf{x}^\top A\mathbf{x}$ equals its own transpose, and transposing flips the sign of A:

$$\mathbf{x}^\top A\mathbf{x} \;=\; \left(\mathbf{x}^\top A\mathbf{x}\right)^\top \;=\; \mathbf{x}^\top A^\top \mathbf{x} \;=\; -\,\mathbf{x}^\top A\mathbf{x} \qquad \Longrightarrow \qquad \mathbf{x}^\top A\mathbf{x} = 0$$

So $Q(\mathbf{x}) = \mathbf{x}^\top S\mathbf{x} + \mathbf{x}^\top A\mathbf{x} = \mathbf{x}^\top S\mathbf{x}$. The quadratic form sees only the symmetric half, and different matrices with the same S give one and the same Q:

matrix	$Q(3,1)$	$Q(1,-1)$	$Q(2,5)$
$G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, the head	16	0	49
$G^\top = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, the head with query and key swapped	16	0	49
$S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, its symmetric half	16	0	49
$\left(\begin{smallmatrix} 1 & 7 \\ -5 & 1 \end{smallmatrix}\right) = S + 6A$	16	0	49

The head and its transpose score every pair in opposite directions, 24 against 8, and their quadratic forms are identical. Whatever a quadratic form records, it is not direction.

Polarization, Recalled: Lengths Give Similarities

The Signatures session derived, for any symmetric inner product, that the squared length of a sum contains the similarity:

$$Q(\mathbf{x}+\mathbf{y}) = Q(\mathbf{x}) + 2\,\mathbf{x}\cdot\mathbf{y} + Q(\mathbf{y}) \qquad \Longrightarrow \qquad \mathbf{x}\cdot\mathbf{y} \;=\; \tfrac{1}{2}\big[\,Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})\,\big]$$

inner product	**vectors**	$Q(\mathbf{x}),\ Q(\mathbf{y}),\ Q(\mathbf{x}+\mathbf{y})$	**polarized**
dot product	$(3,1),\ (1,2)$	$10,\ 5,\ 25$	$\tfrac{1}{2}(25-15) = 5 = 3+2$
declared $(+1,-1,0)$	$(3,1,5),\ (1,1,7)$	$8,\ 0,\ 12$	$\tfrac{1}{2}(12-8) = 2 = 3-1$
the head, $(x_1+x_2)^2$	$(3,1),\ (1,3)$	$16,\ 16,\ 64$	$\tfrac{1}{2}(64-32) = \mathbf{16}$

In the third row the polarized value is $\mathbf{x}_1^{\top} S \mathbf{x}_2 = 16$, the symmetric compatibility of the two tokens. The 16 comes back from Q alone. The 8 does not, and no formula in Q could return it: G and $G^{\top}$ have the same Q and opposite A.

So a quadratic form is a symmetric bilinear form seen along the diagonal, and the diagonal is enough to rebuild the whole symmetric half. This is the **polarization identity**, and it is the hinge of Part III.

Why Polarization Works for Any Matrix

Expand $Q(\mathbf{x}+\mathbf{y})$ with any G, symmetric or not. The cross terms arrive in *both* orders:

$$Q(\mathbf{x}+\mathbf{y}) \;=\; \mathbf{x}^\top G\mathbf{x} \;+\; \underbrace{\mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x}}_{=\,2\,\mathbf{x}^\top S\,\mathbf{y}} \;+\; \mathbf{y}^\top G\mathbf{y}$$

Adding a score to its query-key-swapped copy is the half-sum of Session 11, doubled: the antisymmetric parts cancel and the symmetric parts add. With the head, $24 + 8 = 32 = 2 \cdot 16$.

The same information, two ways

A quadratic form determines exactly one symmetric bilinear form, by polarization, and a symmetric bilinear form determines its quadratic form, by putting $\mathbf{x}$ in both slots. Declaring one is declaring the other. That is why the Signatures session could declare only squares of axes and still fix every similarity.

The $\frac{1}{2}$ needs division by 2, which the real numbers allow; over arithmetic modulo 2 the two notions come apart, a case this course never needs.

What Counts as a Quadratic Form

A quadratic form is any function of the shape $Q(\mathbf{x}) = \mathbf{x}^\top S\mathbf{x}$ with S symmetric; equivalently, a polynomial in the coordinates in which every term has degree exactly 2. Two quick tests: scaling $\mathbf{x}$ by c scales Q by c^2, and the polarization $Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})$ is bilinear.

function	**quadratic form?**	**why**
$x_1^2 + x_2^2$	yes	$S = I$: squared length
$x_1^2 - x_2^2$	yes	$S = \mathrm{diag}(1,-1)$: a declaration
$(x_1 + x_2)^2$	yes	$S = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$: the head
$x_1^2 + 4x_1x_2 + x_2^2$	yes	$S = \left(\begin{smallmatrix}1&2\\2&1\end{smallmatrix}\right)$: Session 10's indefinite form
$\sqrt{x_1^2 + x_2^2}$	no	scales by $\lvert c\rvert$, not c^2
$x_1^2 + x_2$	no	a degree-1 term
$\lvert x_1\rvert + \lvert x_2\rvert$	no	scales by $\lvert c\rvert$; polarization not bilinear

The length is not a quadratic form; its square is. That is why squared lengths, not lengths, carry the algebra, and why the Signatures session declared squares.

Reading the Matrix Off the Polynomial

Every quadratic form is a polynomial with only degree-2 terms. Its symmetric matrix is read off directly: the coefficients of the squares go on the diagonal, and half of each cross coefficient goes in each off-diagonal cell.

$$Q(\mathbf{x}) \;=\; \sum_{i} S_{ii}\, x_i^2 \;+\; \sum_{i<j} 2\, S_{ij}\, x_i x_j$$

polynomial	**symmetric matrix**	**a few values**
$x_1^2 + 4x_1x_2 + x_2^2$	$\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$	$Q(1,1) = 6,\;\; Q(1,-1) = -2,\;\; Q(1, \sqrt{3} - 2) = 0$
$3x_1^2 - 2x_1x_2 + 5x_2^2$	$\left(\begin{smallmatrix} 3 & -1 \\ -1 & 5 \end{smallmatrix}\right)$	$Q(1,0) = 3,\;\; Q(1,1) = 6$
x_1x_2	$\left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$	$Q(1,1) = 1,\;\; Q(1,-1) = -1$

Many matrices produce the same polynomial, since adding any antisymmetric A changes nothing, but exactly one of them is symmetric. That one is what a quadratic form *is*, as a matrix; the head's G and S are two matrices for the one polynomial $(x_1 + x_2)^2$.

The first row is Session 10's indefinite form, signature $(1, 1, 0)$: positive, negative, and zero values from one polynomial.

Part II
Quadratic Forms at Work

Mahalanobis: Length Measured Against the Data

Given data with mean $\boldsymbol{\mu}$ and covariance Σ, measure how far a point lies from the data with the quadratic form of Σ^{-1}:

$$Q_\Sigma(\mathbf{x}) \;=\; (\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})$$

With $\Sigma = I$ this is squared Euclidean distance; the Signatures session used $\Sigma = \mathrm{diag}(4,1)$, an ellipse. Take instead Session 10's correlated matrix, $\Sigma = \left(\begin{smallmatrix} 1 & 0.8 \\ 0.8 & 1 \end{smallmatrix}\right)$, so $\Sigma^{-1} \approx \left(\begin{smallmatrix} 2.78 & -2.22 \\ -2.22 & 2.78 \end{smallmatrix}\right)$:

offset from the mean	**Euclidean, squared**	**Mahalanobis, squared**
$(1,\ 1)$, along the correlation	2	1.11
$(1,\ -1)$, against it	2	10.00

Same Euclidean distance, yet the second point is three times as far (nine times, squared), because it breaks the pattern the data follows; the off-diagonal entries of Σ^{-1} make the axes interact.

In NLP: Podolskiy et al. (2021) fit class means and a shared covariance to transformer embeddings of in-domain utterances, then flag out-of-domain intents by this distance, and report new state-of-the-art results on all the datasets they tested.

The Graph Laplacian: How Much a Labeling Disagrees

On a graph, the Laplacian $L = D - W$ has a quadratic form that sums squared disagreements across edges, $\mathbf{x}^\top L\,\mathbf{x} = \sum_{(i,j)\,\in\,\text{edges}} (x_i - x_j)^2$. The Signatures session used a three-word chain; here is a five-word one:



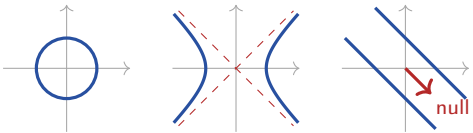
labeling	good	decent	okay	mediocre	bad	$\mathbf{x}^\top L\mathbf{x}$
abrupt	1	1	1	-1	-1	4.00
smooth	1	0.5	0	-0.5	-1	1.00
constant	1	1	1	1	1	0.00

Label propagation clamps seed words (good $= +1$, bad $= -1$) and minimizes this quadratic form over the rest. The minimum is the smooth row: okay lands at 0, neutral. Rao and Ravichandran (2009) induce polarity lexicons this way, over WordNet synonym graphs.

The constant labeling costs nothing: the form measures disagreement, never level. The level is the null axis of this form, and it is why seeds are needed.

Signatures, Recalled

The Signatures session: bring a symmetric S to its eigen-axes, where Q is a sum of signed squares, and count the signs. The counts (p, q, r) do not depend on the axes (Sylvester's law), and the picture of $Q = 1$ shows them:



$x_1^2 + x_2^2$: a circle $(2, 0, 0)$ $x_1^2 - x_2^2$: a hyperbola $(1, 1, 0)$ $(x_1 + x_2)^2$: two lines, $(1, 0, 1)$

quadratic form	**eigenvalues of S**	**signature**
squared Euclidean length on $\mathbb{R}^n$	all 1	$(n, 0, 0)$
Mahalanobis, correlated Σ above	$0.56,\ 5.00$	$(2, 0, 0)$: a tilted ellipse
the five-word chain's Laplacian	$0,\ 0.38,\ 1.38,\ 2.62,\ 3.62$	$(4, 0, 1)$: the level is null
Lorentz, $-x_0^2 + x_1^2 + x_2^2$	$-1,\ 1,\ 1$	$(2, 1, 0)$: hyperbolic embeddings
the head, $(x_1 + x_2)^2$	$2,\ 0$	$(1, 0, 1)$

The Head's Quadratic Form Is Degenerate

The head's $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has eigenvalues 2, along $(1, 1)$, and 0, along $(1, -1)$: signature $(1, 0, 1)$, with null direction $(1, -1)$. Now look at the two tokens:

$$\mathbf{x}_1 - \mathbf{x}_2 = (2,\ -2), \qquad Q(\mathbf{x}_1 - \mathbf{x}_2) = (2 - 2)^2 = 0$$

The two tokens differ by a null vector. Under the symmetric half they cannot be told apart by any key at all:

$$\mathbf{x}_1^\top S\mathbf{y} \ = \ 4\,(y_1 + y_2) \ = \ \mathbf{x}_2^\top S\mathbf{y} \ \text{ for every } \mathbf{y}, \qquad \cos_S(\mathbf{x}_1, \mathbf{x}_2) = \frac{16}{\sqrt{16 \cdot 16}} = 1$$

Everything that distinguishes token 1 from token 2 for this head is directional contribution, in A, exactly the half its quadratic form cannot see.

Why this matters

Session 11 found that a head with tied weights, $W_Q = W_K$, cannot break a tie between two equally similar tokens. The quadratic form shows how complete the tie is: for the symmetric half, these two tokens are one direction.

In-Class Exercise 1

1. For $G = \left(\begin{smallmatrix} 2 & 3 \\ -1 & 1 \end{smallmatrix}\right)$, write $Q(\mathbf{x}) = \mathbf{x}^\top G \mathbf{x}$ as a polynomial, and give the symmetric matrix that represents it.
2. Using *only* values of that Q, compute the similarity of $(1, 0)$ and $(0, 1)$ by polarization. Check it against your matrix.
3. With the same G, compute the raw scores $\mathbf{x}^\top G \mathbf{y}$ and $\mathbf{y}^\top G \mathbf{x}$ for $\mathbf{x} = (1, 0)$, $\mathbf{y} = (0, 1)$. Which of them does polarization return, and what is lost?
4. $\star$ On the chain good–decent–okay–mediocre–bad, why can the energy $\mathbf{x}^\top L \mathbf{x}$ alone never decide whether okay is positive or negative? What do the seeds change, in terms of the signature?

Six minutes. Item 4 is the starred one.

In-Class Exercise 1: Answers

1. $Q = 2x_1^2 + (3-1)\,x_1x_2 + x_2^2 = 2x_1^2 + 2x_1x_2 + x_2^2$, so $S = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. The 3 and the -1 are averaged; their difference, the antisymmetric part, is gone.

2. $Q(1,1) = 5$, $Q(1,0) = 2$, $Q(0,1) = 1$, so the similarity is $\frac{1}{2}(5-2-1) = 1$, the off-diagonal entry of S.

3. $\mathbf{x}^\top G\mathbf{y} = G_{12} = 3$ and $\mathbf{y}^\top G\mathbf{x} = G_{21} = -1$. Polarization returns their average, 1; the half-difference, 2, is the directional contribution $\mathbf{x}^\top A\mathbf{y}$ with $A = \left(\begin{smallmatrix} 0 & 2 \\ -2 & 0 \end{smallmatrix}\right)$, and it is lost.

4. $\star$ Adding the same constant to every label leaves every difference $x_i - x_j$ unchanged, so the energy cannot see the level: the constant vector is a null direction, and the chain's signature is $(4, 0, 1)$. The seeds fix two values, which removes that null direction; the remaining problem has a unique minimum, with okay at 0.

Part III
From a Quadratic Form to a Product

What Would It Mean to Multiply Two Vectors?

The course has met several ways to combine two vectors into one object. Each decided in advance what kind of object the result is, and each keeps a different amount of the pair:

product	**result**	**size**	**of the pair's two halves, keeps**
dot product $\mathbf{u} \cdot \mathbf{v}$	a number	1	one number of the symmetric half
outer product $\mathbf{u}\mathbf{v}^\top$	a matrix	n^2	both halves, entirely
circular convolution $\mathbf{u} \circledast \mathbf{v}$ (HRR)	a vector	n	part of the symmetric half; none of the antisymmetric
elementwise product $\mathbf{u} \odot \mathbf{v}$ (MAP)	a vector	n	the diagonal of the symmetric half; none of the antisymmetric
wedge $\mathbf{u} \wedge \mathbf{v}$ (the bridge deck)	a bivector	$\binom{n}{2}$	the antisymmetric half, entirely

Session 11 put the first and last together, $\mathbf{u}\mathbf{v} = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \wedge \mathbf{v}$, for the dot product. This part asks a different kind of question: instead of choosing what the result is, state one rule the product must obey, and let the kind of object follow. The rule will mention only the quadratic form, so it can be stated for any declared Q, and both halves will come out of it on their own.

Declare Q, Then Ask for a Product

Take any quadratic form Q: the dot product's, a declared signature's, or the head's. Ask for a product of vectors that mentions nothing but Q:

The specification

Find an associative product of vectors, distributive over addition, such that **every vector squares to its quadratic form**:

$$\mathbf{v}\,\mathbf{v} \;=\; Q(\mathbf{v}) \qquad \text{for every vector } \mathbf{v}$$

That single rule is the whole definition of the **Clifford algebra** $\mathrm{Cl}(V, Q)$, which Clifford introduced in 1878 by combining Grassmann's algebra of the wedge with Hamilton's quaternions.

Notice what is *not* in the specification: no antisymmetric form, no wedge, no orientation. Q is the specification, and everything else is derived from it. Session 13b takes this literally: Q is the interpreter, and products of vectors are the programs it runs.

The left side is a product of two vectors; the right side is a number. A product of two vectors is therefore allowed to be a number plus something else, and the next two slides find out what else.

Polarization, Inside the Product

Apply the rule to a sum, $\mathbf{u}+\mathbf{v}$, and expand by distributivity, keeping the order of each product, since nothing yet says $\mathbf{uv}$ and $\mathbf{vu}$ are equal:

$$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) \;=\; \mathbf{uu}+\mathbf{uv}+\mathbf{vu}+\mathbf{vv} \;=\; Q(\mathbf{u}+\mathbf{v})$$

Replace $\mathbf{uu}$ and $\mathbf{vv}$ by $Q(\mathbf{u})$ and $Q(\mathbf{v})$, which the rule allows, and solve:

$$\mathbf{uv}+\mathbf{vu} \;=\; Q(\mathbf{u}+\mathbf{v})-Q(\mathbf{u})-Q(\mathbf{v}) \;=\; 2\,S(\mathbf{u},\mathbf{v})$$

The right side is the polarization identity of Part I. The symmetric half of the product is forced: $\frac{1}{2}(\mathbf{uv}+\mathbf{vu}) = S(\mathbf{u},\mathbf{v})$, the similarity the declared Q fixes. With the head's form and its two tokens, $(\mathbf{x}_1+\mathbf{x}_2)^2 = 64$ and $\mathbf{x}_1^2 = \mathbf{x}_2^2 = 16$, so $\mathbf{x}_1\mathbf{x}_2+\mathbf{x}_2\mathbf{x}_1 = 32 = 2\cdot 16$.

Squaring a sum produces both orders of the cross term, and adding the two orders is how a quadratic form returns its symmetric half. The same three lines that proved polarization in Part I now fix half of a product.

The Other Half Is Left Free

The rule says nothing about the difference $\mathbf{uv} - \mathbf{vu}$. The algebra keeps it, as an object of a new kind, the **wedge** of the bridge deck:

$$\mathbf{u} \wedge \mathbf{v} \;:=\; \tfrac{1}{2}\,(\mathbf{uv} - \mathbf{vu}), \qquad \boxed{\mathbf{uv} \;=\; S(\mathbf{u},\mathbf{v}) \;+\; \mathbf{u} \wedge \mathbf{v}}$$

half	**determined by**	**kind of object**	**records**
symmetric, $S(\mathbf{u},\mathbf{v})$	the declared Q, by polarization	a number	the declared similarity
antisymmetric, $\mathbf{u} \wedge \mathbf{v}$	the two vectors alone; Q is never consulted	a bivector: an oriented plane	which came first, and the plane they span

Session 11 derived $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ for the ordinary dot product. Nothing in the derivation above used the dot product: the same formula holds for every quadratic form, with S in place of the dot.

This answers the puzzle. The declaration fixes only Q. The antisymmetric half was never declared because it never needed declaring: in $\mathbf{uv} - \mathbf{vu}$ every square $\mathbf{vv}$ cancels before Q is consulted, so the wedge of two vectors does not depend on Q at all.

A Basis Gives the Whole Multiplication Table

Choose axes that are independent for S, so $S(\mathbf{e}_i, \mathbf{e}_j) = 0$ for $i \neq j$. Polarization in the product then gives the two rules that generate everything:

$$\mathbf{e}_i\mathbf{e}_j + \mathbf{e}_j\mathbf{e}_i = 2\,S(\mathbf{e}_i, \mathbf{e}_j) = 0 \;\Rightarrow\; \mathbf{e}_i\mathbf{e}_j = -\,\mathbf{e}_j\mathbf{e}_i \quad (i \neq j), \qquad\qquad \mathbf{e}_i\mathbf{e}_i = Q(\mathbf{e}_i)$$

The first rule is swap: different axes trade places at the cost of a sign. The second is collapse: an axis times itself becomes its declared square. By Sylvester the axes can be scaled so that each $Q(\mathbf{e}_i)$ is $+1$, -1, or 0, and with p, q and r of each the algebra is $\mathrm{Cl}(p, q, r)$: the signature of the form is the name of the algebra it generates.

The head's own axes. $\mathbf{g}_1 = (\frac{1}{2}, \frac{1}{2})$ and $\mathbf{g}_2 = (1, -1)$ are independent for the head's S, since $S(\mathbf{g}_1, \mathbf{g}_2) = (\frac{1}{2} + \frac{1}{2})(1 - 1) = 0$, with $Q(\mathbf{g}_1) = 1$ and $Q(\mathbf{g}_2) = 0$. The head's algebra is $\mathrm{Cl}(1, 0, 1)$, and its table is three lines:

$$\mathbf{g}_1\mathbf{g}_1 = 1, \qquad \mathbf{g}_2\mathbf{g}_2 = 0, \qquad \mathbf{g}_1\mathbf{g}_2 = -\,\mathbf{g}_2\mathbf{g}_1$$

The next slide multiplies the two tokens with nothing but these three lines.

The Head's Two Halves, Recomputed in Its Own Axes

In the head's axes the two tokens are $\mathbf{x}_1 = 4\mathbf{g}_1 + \mathbf{g}_2 = (3, 1)$ and $\mathbf{x}_2 = 4\mathbf{g}_1 - \mathbf{g}_2 = (1, 3)$. Multiply, using swap and collapse:

$$\mathbf{x}_1\mathbf{x}_2 = 16\,\mathbf{g}_1\mathbf{g}_1 - 4\,\mathbf{g}_1\mathbf{g}_2 + 4\,\mathbf{g}_2\mathbf{g}_1 - \mathbf{g}_2\mathbf{g}_2 = 16 - 4\,\mathbf{g}_1\mathbf{g}_2 - 4\,\mathbf{g}_1\mathbf{g}_2 - 0 = 16 - 8\,\mathbf{g}_1\mathbf{g}_2$$
$$\mathbf{x}_2\mathbf{x}_1 = 16\,\mathbf{g}_1\mathbf{g}_1 + 4\,\mathbf{g}_1\mathbf{g}_2 - 4\,\mathbf{g}_2\mathbf{g}_1 - \mathbf{g}_2\mathbf{g}_2 = 16 + 8\,\mathbf{g}_1\mathbf{g}_2$$

	the head's score	**the Clifford product**
symmetric half, $1 \to 2$ and $2 \to 1$	$16,\ \ 16$	$16,\ \ 16$: declared, through Q
antisymmetric half, $1 \to 2$ and $2 \to 1$	$+8,\ \ -8$	$-8\,\mathbf{g}_1\mathbf{g}_2,\ \ +8\,\mathbf{g}_1\mathbf{g}_2$: computed

The head had to learn both halves. The algebra was told only the first, and produced the second as a bivector with coefficient 8. The sign depends on which way round the unit bivector is taken: $\mathbf{g}_1\mathbf{g}_2 = -\,\mathbf{e}_1 \wedge \mathbf{e}_2$, so $-8\,\mathbf{g}_1\mathbf{g}_2 = +8\,\mathbf{e}_1 \wedge \mathbf{e}_2$, and in the standard orientation the two columns agree exactly.

Also, $(\mathbf{x}_1 - \mathbf{x}_2)^2 = (2\mathbf{g}_2)^2 = 0$: the null difference of Part II, now as a product that collapses to zero.

One Word, Three Values: Reducing Products by the Two Moves

The outer product never forgets: $\mathbf{e}_2 \otimes \mathbf{e}_1$, $\mathbf{e}_2 \otimes \mathbf{e}_1 \otimes \mathbf{e}_2$, and so on are all distinct objects, and products of k vectors need n^k coordinates, the growth of Session 11b and the cost of TPR.

The Clifford rules cut this down. Any word in the axes reduces by the two moves, swap neighbors ($\mathbf{e}_j\mathbf{e}_i = -\mathbf{e}_i\mathbf{e}_j$) and collapse equal neighbors ($\mathbf{e}_i\mathbf{e}_i = Q(\mathbf{e}_i)$). Take the word $\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2\mathbf{e}_1$:

declared signature	reduction	value
$\mathrm{Cl}(2,0,0)$: $\mathbf{e}_1^2 = \mathbf{e}_2^2 = +1$	$\mathbf{e}_2\mathbf{e}_1\,(\mathbf{e}_2\mathbf{e}_2)\,\mathbf{e}_1 = \mathbf{e}_2\,(\mathbf{e}_1\mathbf{e}_1)$	$\mathbf{e}_2$
$\mathrm{Cl}(1,1,0)$: $\mathbf{e}_2^2 = -1$	$\mathbf{e}_2\mathbf{e}_1\,(-1)\,\mathbf{e}_1 = -\,\mathbf{e}_2\,(\mathbf{e}_1\mathbf{e}_1)$	$-\mathbf{e}_2$
$\mathrm{Cl}(1,0,1)$: $\mathbf{e}_2^2 = 0$	$\mathbf{e}_2\mathbf{e}_1\,(0)\,\mathbf{e}_1$	0

One word, three values. The outer product cannot evaluate the word at all; the Clifford algebra can, and what it returns depends on nothing but the declared quadratic form. Session 13b starts from this slide: the two moves are the interpreter, and a word is a program.

Every word reduces to $\pm$ a sorted product of *distinct* axes, or to 0. Formally, $\mathrm{Cl}(V, Q)$ is the algebra of all words with $\mathbf{v}\mathbf{v} - Q(\mathbf{v})$ set to zero for every $\mathbf{v}$.

Tensors Grow; the Clifford Algebra Stays Put

A reduced word is a choice of *which* axes appear: 2^n of them, sorted by **grade**, the number of axes in the word: a number has grade 0, a vector 1, a bivector 2.

$n = 4$, degree or grade k	0	1	2	3	4	**total**
tensor products, n^k	1	4	16	64	256	keeps growing
Clifford grade k, $\binom{n}{k}$	1	4	6	4	1	$16 = 2^4$

A product of two vectors lands in grades 0 and 2: the symmetric half in grade 0, the antisymmetric half in grade 2, each in a place of its own. At $n = 300$, grades 0 through 2 hold $1 + 300 + 44{,}850 = 45{,}151$ coordinates, against $90{,}000$ for a 300×300 matrix. The saving has a price: grade 2 is alternating, so it drops the symmetric and diagonal structure that a general relation needs (13c).

	size	**antisymmetric half**	**unbinding**
TPR	grows as n^k	kept	exact, with orthonormal roles
VSA (HRR, MAP)	fixed, d	folded away	approximate; needs cleanup
Clifford algebra	fixed, 2^n, graded	kept, in grade 2	exact, with orthonormal roles

Part IV
The Smallest Algebras

what one or two declared letters generate

The Coefficients Stay Real; the New Letter Is a Direction

One distinction first. The symbol $\mathbb{R}$ plays two different parts:

As the scalars

Numbers such as 2, -3.7, π are **coefficients**. They say how much of a vector there is, and they multiply as numbers always did.

As a one-dimensional vector space

$V = \mathbb{R}\,\mathbf{e}$ has vectors $\mathbf{v} = x\,\mathbf{e}$ with $x \in \mathbb{R}$. Here $\mathbf{e}$ is a **basis direction**, a letter of the alphabet, not a scalar.

The coefficients stay real in everything that follows. The only new question is what $\mathbf{ee}$ should be, and the declaration answers it: $\mathbf{ee} = Q(\mathbf{e})$, with $Q(x\,\mathbf{e}) = Q(\mathbf{e})\,x^2$.

declaration	**quadratic form**	**so**
$\mathbf{e}^2 = +1$	$Q(x\,\mathbf{e}) = +x^2$	$(3\mathbf{e})^2 = 9$
$\mathbf{e}^2 = -1$	$Q(x\,\mathbf{e}) = -x^2$	$(3\mathbf{e})^2 = -9$
$\mathbf{e}^2 = 0$	$Q(x\,\mathbf{e}) = 0$	$(3\mathbf{e})^2 = 0$

A general vector need not square to $+1$, -1 or 0; those are the normalized values for basis letters, as Sylvester's theorem allows.

One Generator, Three Algebras

Start with a single axis $\mathbf{e}$, as at the end of the Signatures session. The algebra has elements $a + b\,\mathbf{e}$, and the declared sign of $\mathbf{e}^2$ is the only choice there is:

$$(a + b\,\mathbf{e})(c + d\,\mathbf{e}) \;=\; \big(ac + bd\,\mathbf{e}^2\big) + (ad + bc)\,\mathbf{e}$$

$\mathbf{e}^2$	**algebra**	**worked**	**known as**
$+1$	$\mathrm{Cl}(1,0,0)$	$(2+3\mathbf{e})(1+4\mathbf{e}) = 14 + 11\mathbf{e}$	split-complex numbers
-1	$\mathrm{Cl}(0,1,0)$	$(2+3\mathbf{e})(1+4\mathbf{e}) = -10 + 11\mathbf{e}$	the complex numbers
0	$\mathrm{Cl}(0,0,1)$	$(3+\mathbf{e})(3+\mathbf{e}) = 9 + 6\mathbf{e}$	dual numbers

One sign, three different worlds. Session 13b shows what each world is good for: $+1$ splits things into two exclusive pieces, -1 turns them in a cycle, and 0 carries a rate of change.

The complex numbers are not postulated here. They are the Clifford algebra of a single direction declared to have squared length -1.

Squaring in Each Algebra

Keep the multiplication algorithm fixed, $(a + b\,\mathbf{e})(c + d\,\mathbf{e}) = (ac + bd\,\mathbf{e}^2) + (ad + bc)\,\mathbf{e}$, and change only the line that says what $\mathbf{e}^2$ is. Square $2 + 3\mathbf{e}$ three times:

declaration	**expansion**	**result**	**known as**
$\mathbf{e}^2 = -1$, write $\mathbf{e} = i$	$4 + 12i + 9i^2 = 4 + 12i - 9$	$-5 + 12i$	complex numbers
$\mathbf{e}^2 = +1$, write $\mathbf{e} = j$	$4 + 12j + 9j^2 = 4 + 12j + 9$	$13 + 12j$	split-complex numbers
$\mathbf{e}^2 = 0$, write $\mathbf{e} = \varepsilon$	$4 + 12\varepsilon + 9\varepsilon^2 = 4 + 12\varepsilon + 0$	$4 + 12\varepsilon$	dual numbers

Only the $9\,\mathbf{e}^2$ term differs, and it differs by nothing but the declared sign. Same coefficients, different squaring rule, different algebra.

In the complex case there is no mysterious operation: the real scalars were enlarged by one direction, and that direction was told what its square is. The -9 is the declaration showing up inside a computation.

The dual-number row is the one in which the second-order term vanishes. Session 13b shows that this is what makes ε carry a derivative.

Two Generators: The Sign of the Bivector

With two orthogonal basis vectors, the bivector $\mathbf{e}_1\mathbf{e}_2$ appears, and its square is decided by the declared signs:

$$(\mathbf{e}_1\mathbf{e}_2)^2 \;=\; \mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2 \;=\; -\,\mathbf{e}_1\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2 \;=\; -\,Q(\mathbf{e}_1)\,Q(\mathbf{e}_2)$$

signature	$\mathbf{e}_1^2,\ \mathbf{e}_2^2$	$(\mathbf{e}_1\mathbf{e}_2)^2$	**what lives there**
$\mathrm{Cl}(2,0,0)$	$+1,\ +1$	-1	rotations of the plane; the complex numbers inside it
$\mathrm{Cl}(1,1,0)$	$+1,\ -1$	$+1$	boosts, as in special relativity (Session 13b)
$\mathrm{Cl}(0,2,0)$	$-1,\ -1$	-1	the quaternions
$\mathrm{Cl}(1,0,1)$	$+1,\ 0$	0	translations (Session 13b)

Quaternions, for free. In $\mathrm{Cl}(0,2,0)$ put $i = \mathbf{e}_1$, $j = \mathbf{e}_2$, $k = \mathbf{e}_1\mathbf{e}_2$. Then $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, and $ijk = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2 = -\mathbf{e}_1\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2 = -(-1)(-1) = -1$: Hamilton's rules of 1843 are the multiplication table of a declared quadratic form with two minus signs. Session 14 finds the same rules among the planes of ordinary space.

A Small Family of Declarations

letters declared	squares	algebra	known as	dim.
none	–	$\mathrm{Cl}(0,0,0)$	the real numbers	1
one negative	$\mathbf{e}^2 = -1$	$\mathrm{Cl}(0,1,0)$	the complex numbers	2
one positive	$\mathbf{e}^2 = +1$	$\mathrm{Cl}(1,0,0)$	split-complex numbers	2
one null	$\mathbf{e}^2 = 0$	$\mathrm{Cl}(0,0,1)$	dual numbers	2
two negative	$\mathbf{e}_1^2 = \mathbf{e}_2^2 = -1$	$\mathrm{Cl}(0,2,0)$	the quaternions	4
two positive	$\mathbf{e}_1^2 = \mathbf{e}_2^2 = +1$	$\mathrm{Cl}(2,0,0)$	the plane; even part: the complex numbers	4
three positive	all $+1$	$\mathrm{Cl}(3,0,0)$	space; even part: the quaternions (Session 14)	8

Two routes to the complex numbers: one letter declared -1, or the even programs $a + b\,\mathbf{e}_1\mathbf{e}_2$ of the Euclidean plane, where the bivector plays i. In the second, i is an oriented plane element, not an added scalar direction. That is the reading Session 13b develops.

The quadratic form does not merely report geometry. Once imposed as $\mathbf{v}^2 = Q(\mathbf{v})$, it fixes the multiplication table of the entire algebra, and with it which familiar number system the algebra is.

In-Class Exercise 2

1. In $\mathrm{Cl}(2,0,0)$, compute $(\mathbf{e}_1 + 2\mathbf{e}_2)(3\mathbf{e}_1 - \mathbf{e}_2)$ by swap and collapse, and split the result into its symmetric and antisymmetric halves.
2. Repeat the computation in $\mathrm{Cl}(1,1,0)$, where $\mathbf{e}_2^2 = -1$. Which half changed, and why that one?
3. Multiply $(2 + 3\mathbf{e})(1 + 4\mathbf{e})$ under each of the three one-letter declarations.
4. $\star$ In $\mathrm{Cl}(2,0,0)$ the bivector $\mathbf{e}_1\mathbf{e}_2$ squares to -1, and in $\mathrm{Cl}(0,2,0)$ so does the letter $\mathbf{e}_1$. Both algebras contain a copy of the complex numbers. What distinguishes them?

Seven minutes. Item 2 is the one to discuss.

In-Class Exercise 2: Answers

1. $3\mathbf{e}_1\mathbf{e}_1 - \mathbf{e}_1\mathbf{e}_2 + 6\mathbf{e}_2\mathbf{e}_1 - 2\mathbf{e}_2\mathbf{e}_2 = 3 - \mathbf{e}_1\mathbf{e}_2 - 6\mathbf{e}_1\mathbf{e}_2 - 2 = 1 - 7\,\mathbf{e}_1\mathbf{e}_2$. Symmetric half 1 (the dot product, $3-2$); antisymmetric half $-7\,\mathbf{e}_1\mathbf{e}_2$ (the wedge, $1\cdot(-1) - 2\cdot 3$).

2. Now $\mathbf{e}_2\mathbf{e}_2 = -1$, so the result is $3 + 2 - 7\,\mathbf{e}_1\mathbf{e}_2 = 5 - 7\,\mathbf{e}_1\mathbf{e}_2$. Only the symmetric half changed: it is the polarization of Q, and Q is what was changed. The wedge does not depend on Q, so it stayed at -7.

3. $(ac + bd\,\mathbf{e}^2) + (ad + bc)\,\mathbf{e}$ with $a, b, c, d = 2, 3, 1, 4$: $\mathbf{e}^2 = +1$ gives $14 + 11\mathbf{e}$; $\mathbf{e}^2 = -1$ gives $-10 + 11\mathbf{e}$; $\mathbf{e}^2 = 0$ gives $2 + 11\mathbf{e}$. The $\mathbf{e}$ coefficient never sees the declaration.

4. ⋆ The copies of the complex numbers agree: $a + b\,\mathbf{e}_1\mathbf{e}_2$ and $a + b\,\mathbf{e}_1$ multiply alike. The algebras differ in their *vectors*: in $\mathrm{Cl}(2,0,0)$ the letters square to $+1$ and can be mirrors (Session 13b), while in $\mathrm{Cl}(0,2,0)$ they square to -1 and are themselves imaginary units, i and j. One algebra is four numbers holding a plane; the other is four numbers holding the quaternions.

Where Each Ingredient Comes From: The Next Two Sessions

declared, today	**computed by the algebra**	**added by FGA**	**where**
the quadratic form Q	S by polarization: the symmetric half	a signature chosen for meaning	13b
–	$\mathbf{u} \wedge \mathbf{v}$: the antisymmetric half	subspaces for roles and slots	13c
an orthogonal basis	$\mathrm{Cl}(p, q, r)$: the whole table	grade and subspace as type	13c
–	inverses, mirrors, rotors	wedge binding; exact unbinding	13b, 13c
–	projectors, cycles, rates	polar pairs; coercion by rotor	13b

Read the first column: the only thing declared is a quadratic form. Everything in the second column follows from $\mathbf{v}^2 = Q(\mathbf{v})$: today derived the first three rows, and 13b derives the rest. The third column is what Functional Geometric Algebra builds on top, in 13b and 13c.

Session 10 asked what the matrix in the middle of $\mathbf{x}^\top G \mathbf{y}$ should be, and Session 11 split it in two. The Clifford algebra is the answer that declares one half and derives the other.

Summary

1. A **quadratic form** is a bilinear form with the same vector in both slots, $Q(\mathbf{x}) = \mathbf{x}^\top G\mathbf{x}$. Since $\mathbf{x}^\top A\mathbf{x} = 0$, it sees only the symmetric half.
2. **Polarization**, $S(\mathbf{x}, \mathbf{y}) = \frac{1}{2}[Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})]$, recovers that half exactly; nothing recovers the other, since G and $G^\top$ have one Q.
3. Quadratic forms are everywhere in NLP: Mahalanobis distance for out-of-domain detection, the Laplacian energy of label propagation, the Lorentz form of hyperbolic embeddings.
4. A quadratic form is classified by its **signature** (p, q, r), read off the eigenvalues of S. The head of Session 11 has signature $(1, 0, 1)$; its two tokens differ by a null vector.
5. The **Clifford algebra** is defined by one rule, $\mathbf{v}\mathbf{v} = Q(\mathbf{v})$. Polarization forces the symmetric half of the product; the wedge comes free: $\mathbf{u}\mathbf{v} = S(\mathbf{u}, \mathbf{v}) + \mathbf{u} \wedge \mathbf{v}$. Swap and collapse evaluate every word; one word can take three values under three declarations.
6. The **smallest algebras** are familiar number systems: one letter declared -1, $+1$ or 0 gives the complex, split-complex and dual numbers; two letters declared -1 give the quaternions.

Coming up: 13b reads Q as an interpreter and products as programs; 13c builds the wedge and the dual and returns to binding; Session 14 gives the full table of $\mathrm{Cl}(3, 0, 0)$.

Exercises

1. Prove that $\mathbf{x}^\top A\mathbf{x} = 0$ for every antisymmetric A, and conclude that no function of Q can recover the antisymmetric half of a form.
2. Show that polarizing $Q(\mathbf{x}) = \mathbf{x}^\top S\mathbf{x}$ returns $S(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top S\mathbf{y}$ for every symmetric S.
3. Find the signatures of the Laplacians of a triangle (three words, all linked) and of two separate edges (four words). Relate the number of zeros to the number of connected components.
4. In $\mathrm{Cl}(3,0,0)$, show that $(\mathbf{e}_1\mathbf{e}_2)^2 = (\mathbf{e}_2\mathbf{e}_3)^2 = (\mathbf{e}_3\mathbf{e}_1)^2 = -1$. Compute $(\mathbf{e}_2\mathbf{e}_3)(\mathbf{e}_3\mathbf{e}_1)$, and find signs for i, j, k that make these bivectors obey $ij = k$.
5. Reduce the word $\mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_3\mathbf{e}_2$ by swap and collapse in $\mathrm{Cl}(3,0,0)$, then in $\mathrm{Cl}(2,1,0)$ with $\mathbf{e}_3^2 = -1$, then in $\mathrm{Cl}(2,0,1)$ with $\mathbf{e}_3^2 = 0$.
6. $\star$ Show that in $\mathrm{Cl}(2,0,0)$ the four elements $1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1\mathbf{e}_2$ are closed under the product, and write out the full 4×4 table. Which entries would change in $\mathrm{Cl}(1,1,0)$, and which would not?

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